Henry Hsu

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WORK EXPERIENCE

Software Engineering Intern **Jun. 2026 – Aug. 2026**

Google *Taipei, Taiwan*

- **Test Infrastructure:** Built **3** integration test infrastructures from a **zero** baseline (**Java**, **Python**), onboarding **400+** test suites across **200+** smart home device types, with **90%+** unit test coverage for the infrastructure itself
- **Automated Failure Triage:** Built an LLM-backed service that triages scheduled integration test runs and posts root-cause summaries directly onto issue tickets, replacing manual log inspection and cutting failure narrow-down time by **20+ minutes** per issue
- **Local Reproduction Tooling:** Extended the service to reproduce device failures locally against virtual devices instead of waiting on cloud runs, cutting manual reproduction time by **85%**; **shipped and adopted by the team**

Generative AI Team Lead **Dec. 2024 – Aug. 2025**

MaiAgent Co., Ltd. *– Award-Winning B2B GenAI Startup* *Taipei, Taiwan*

- **API Optimization:** Optimized **140+** RESTful APIs through SQL query refactoring, connection pooling (Elasticsearch, Cohere, OpenAI), and Django caching; cut response time of 13 high-traffic APIs by **27.7%** overall and eliminated N+1 queries
- **CI/CD & Testing:** Built a **GitHub Actions** CI pipeline with pytest unit and E2E coverage, reaching **67%** code coverage from a **zero** baseline; reduced production hotfixes by **90%** initially and **50%** long term; managed database migrations in the CD pipeline
- **Open Source:** Contributed **14** merged pull requests to **LlamaIndex** (**15K+ stars**), fixing bugs and adding features across the AWS Bedrock, Claude, Elasticsearch, Cohere, OpenAI, and MCP Client integrations

Generative AI Intern **Sep. 2024 – Dec. 2024**

MaiAgent Co., Ltd. *– Award-Winning B2B GenAI Startup* *Taipei, Taiwan*

- **Data Pipeline:** Built an async indexing pipeline with **asyncio** and **Celery**, **3.5x faster** parsing while scaling **3M to 20M+** text chunks
- **Real-Time Communication:** Implemented a **WebSocket** notification system with **Redis** pub/sub for file parsing status; built event-driven architecture for agent state transitions, pushing render triggers to the frontend for live UI updates

EDUCATION

Rice University - Top 20 U.S. University **Aug. 2025 – Dec. 2026 (Expected)**

M.C.S., Computer Science (GPA: 4.00/4.00) *Houston, TX*

National Yang Ming Chiao Tung University (NYCU) - Top 3 Taiwan University **Sep. 2020 – Jun. 2024**

B.S., Industrial Engineering and Management (GPA: 4.07/4.30, Two-Time Dean's List) *Hsinchu, Taiwan*

Minor: Computer Science (Domain GPA: 4.13/4.30)

PROJECTS

Prompt-Aware Image Token Pruning for Multimodal Inference | *PyTorch, LLaVA, CLIP ViT, Knowledge Distillation* **Feb. 2026 – May 2026**

- **Inference Efficiency:** Distilled a **1.3M-parameter** scorer from LLaVA-1.5-7B attention to drop the image patches the model ignores before the projector, cutting attention compute for the frozen LLM **10**× and running **150**× cheaper than a CLIP-GradCAM pass
- **Accuracy & Latency:** Kept a quarter of the image tokens and still answered TextVQA about as well as the full model, **12.3** points clear of random pruning, while deciding what to keep in **0.43 ms** an image, cheap enough to be free at inference

Full-Stack Social Network (Ricebook) | *React, Redux Toolkit, Express, MongoDB, OAuth 2.0, Vitest* **Sep. 2025 – Dec. 2025**

- **Full-Stack Build:** Built a **React 19** single-page app on Redux Toolkit against an **Express 5**/MongoDB API, with session-cookie auth and **Google OAuth 2.0** linking; covered the Redux store with **30** Vitest suites at **97%** statement coverage

Relational Database Engine from Scratch | *C++, B+ Tree, Query Optimization, TPC-H* **Aug. 2025 – Dec. 2025**

- **Storage Engine:** Built the C++ persistence layer, a buffer manager with **LRU** eviction and page pinning, slotted-page record storage, and a **B+ tree** index supporting range scans
- **Query Processing:** Implemented the relational operators, including a **sort-merge join** over external merge sort that spills to disk, under a SQL compiler and a cost-based optimizer that orders joins from cardinality statistics; ran **TPC-H** at scale factor 1

AWARDS

Presidential Hackathon Winner (2024, Top 5 Nationally) - Urban noise agent: structured-output LLM pipeline and tool calling

23rd Golden Peak Award (2025) - Outstanding Commercial Product, MaiAgent AI Platform

SKILLS

Languages: Python, JavaScript, TypeScript, Java, C/C++ | **Frontend:** React, HTML, CSS | **Backend:** Django, FastAPI, Express.js
GenAI: LlamaIndex, LangChain, LangGraph, MCP | **Databases:** PostgreSQL, MongoDB, Elasticsearch | **Async:** asyncio, Celery
APIs: RESTful, WebSocket, Redis Pub/Sub | **Testing:** pytest, Vitest, JUnit | **DevOps:** GitHub Actions, Docker, git, Nginx | **ML:** PyTorch